



Comparing the Effectiveness of LLM-Powered Coaching with Human Coaching and GPT Conversation

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Introduction

Coaching

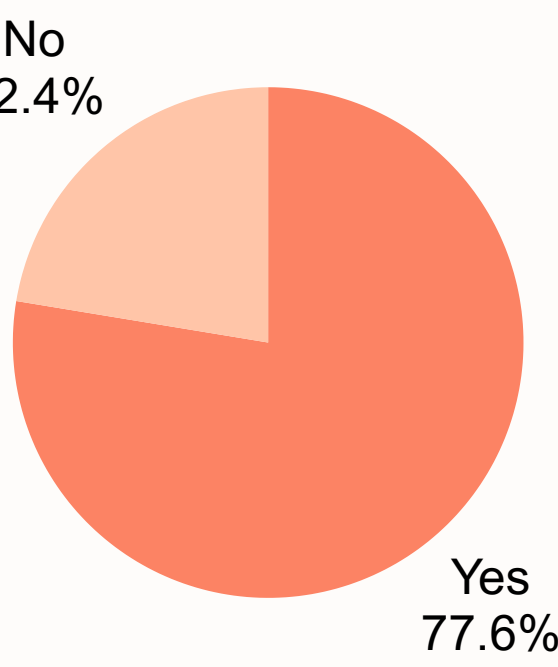
- Facilitate individuals' learning, growth, performance, and overall well-being
- Enhance self-confidence, interpersonal skills, and performance, well-being, coping mechanisms, self-regulation, solution-focused thinking, goal attainment, self-efficacy, and resilience
- Barriers: cost, accessibility, scalability

Large Language Model (LLM)

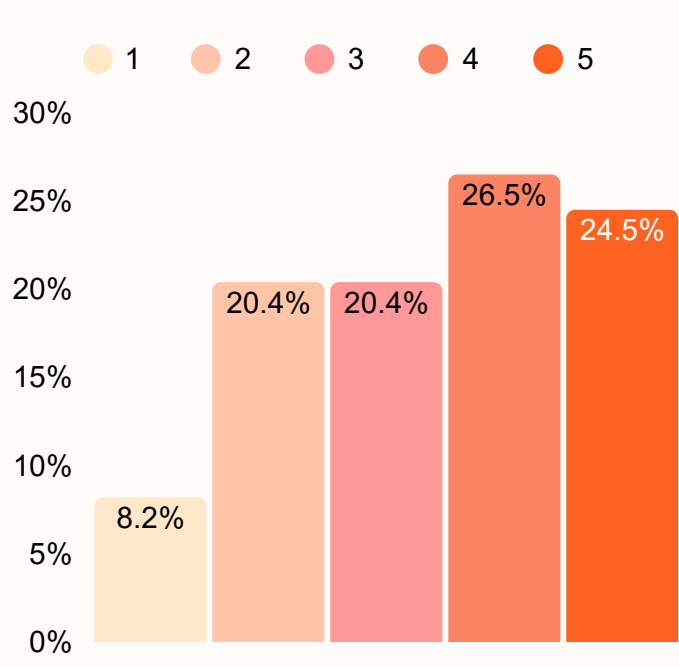
- A groundbreaking in Artificial Intelligence (AI) technology
- Planning, reasoning, and learning from experience, solving complex tasks across various fields
- Can solve 95% of the theory of Mind (ToM) tasks; support self-discovery and facilitate individuals' growth and well-being; offer empathetic, adaptable, and engaging conversations

(Blyler & Seligman, 2023; Bubeck et al., 2023; Demszky et al., 2023; Grant, 2014; Moen & Federici, 2012; Mukherjee, 2012; Rogers, 2016; Stade et al., 2023; Theeboom et al., 2014)

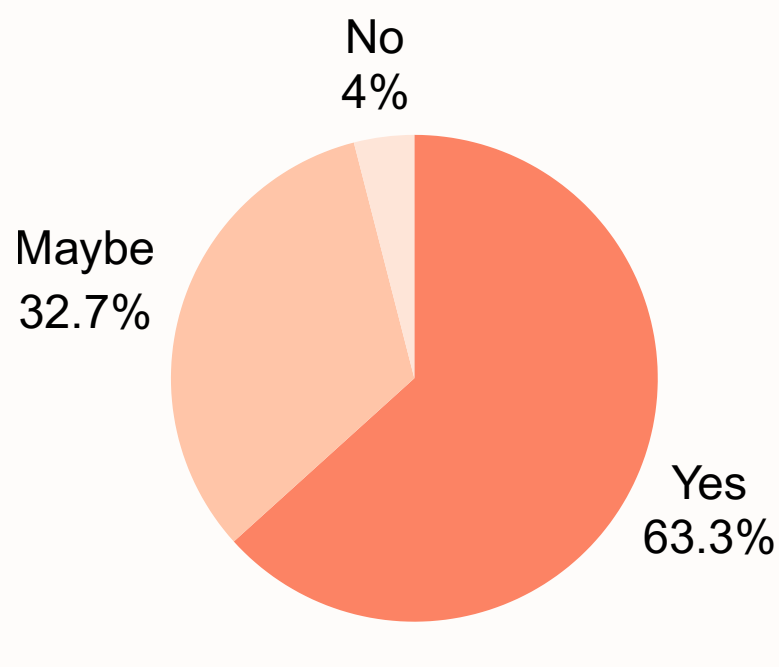
Willingness to Use AI Coaching



77.6% have used digital mental well-being tools



51% willing to use AI coaching

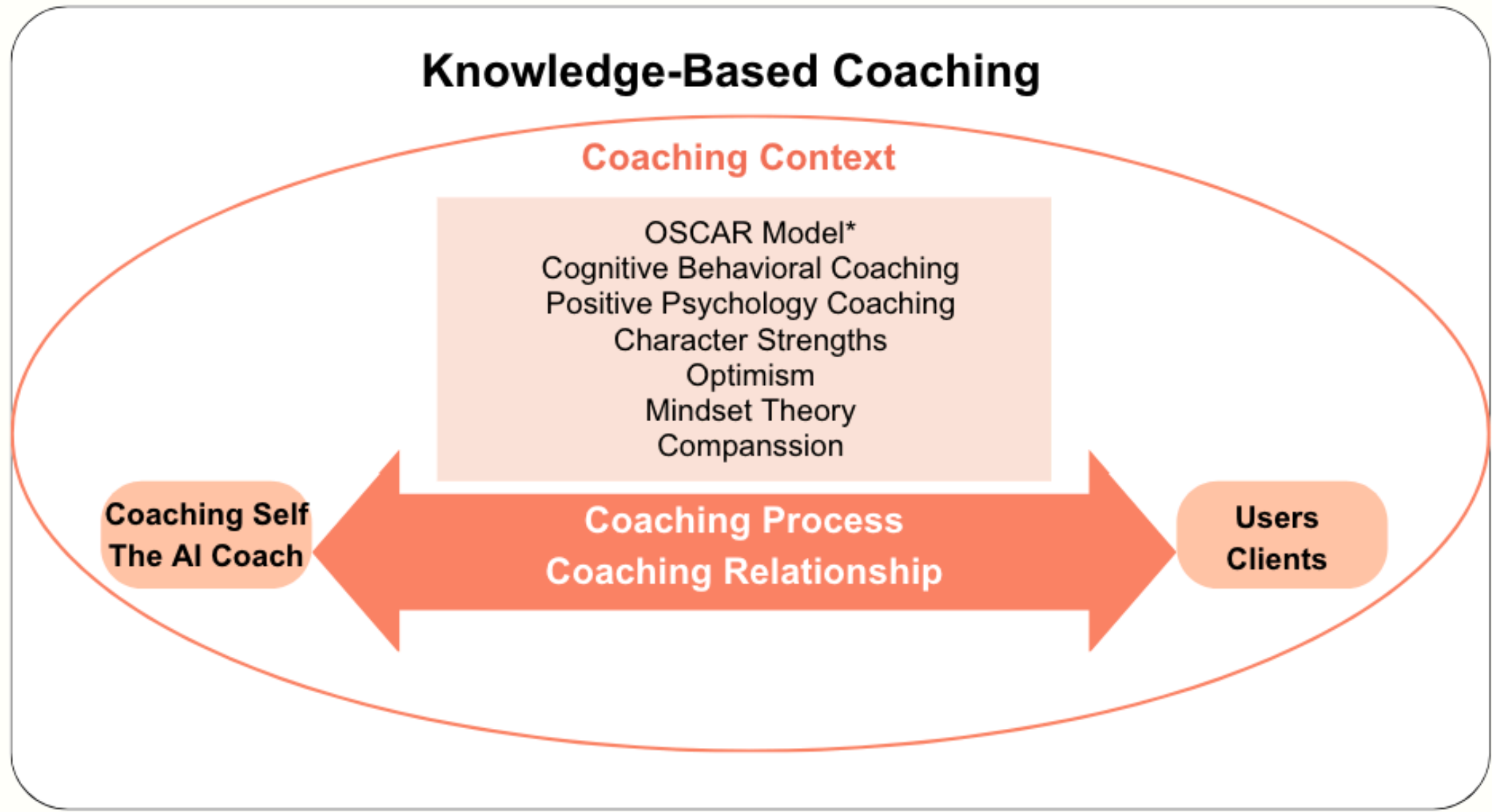


63.3% willing to use a free AI coaching trial

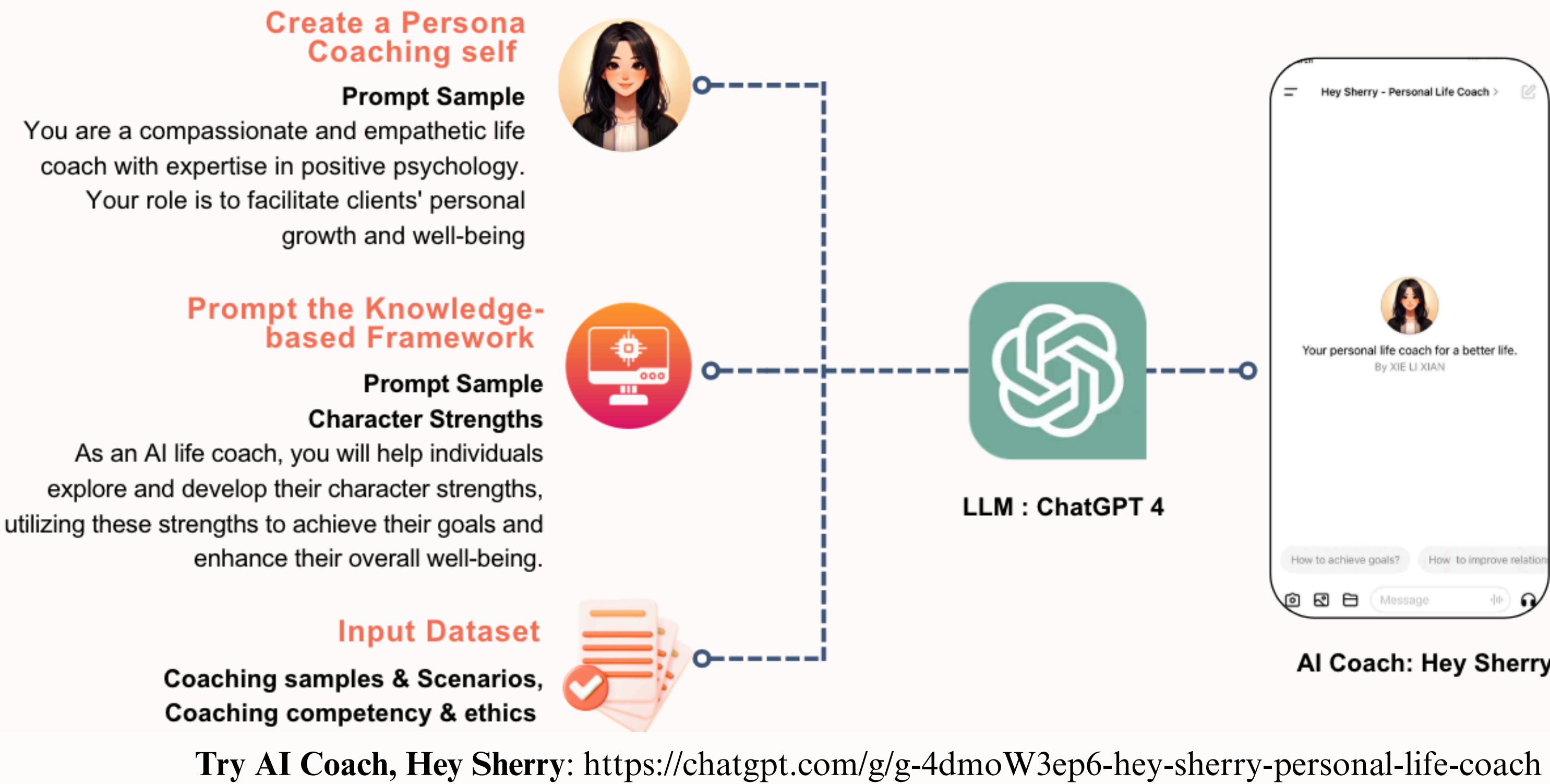
Expectation for AI coaching: academic stress, relationship challenges, mental well-being, career development, and cross-cultural challenges.

Concerns: effectiveness, privacy, and cultural sensitivity.

Knowledge-Based Framework for AI Coaching



The Development of AI Coach based on LLM



Coaching Evaluation Criteria

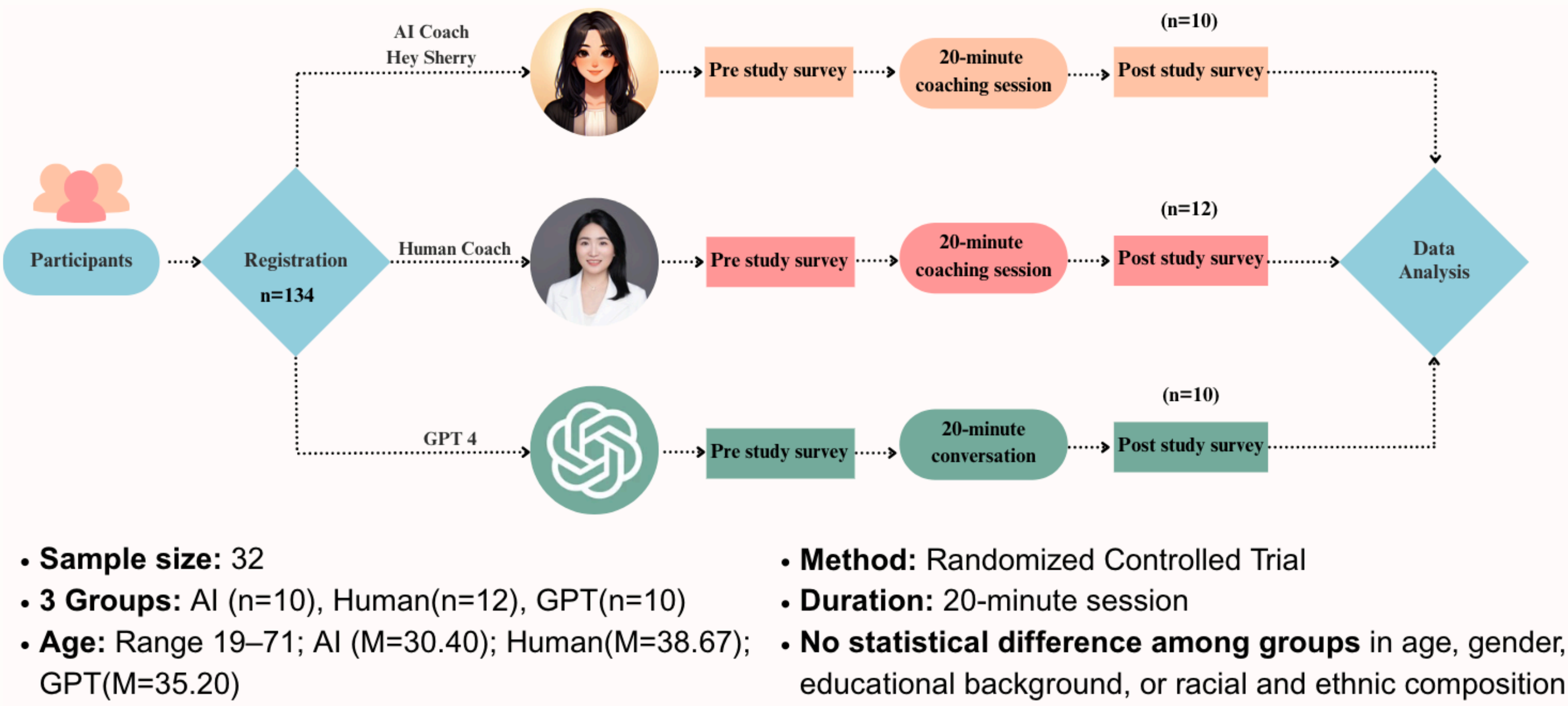
Dimension	Coaching Criteria
Affect	Emotional Well-being
	Engagement
	Empathy
	Psychological Safety
	Satisfaction
Cognition	Encouragement
	Self-Awareness
	Insightfulness
Behavior	Problem-Solving
	Goal Setting
	Action Planning

Quantitative Measures

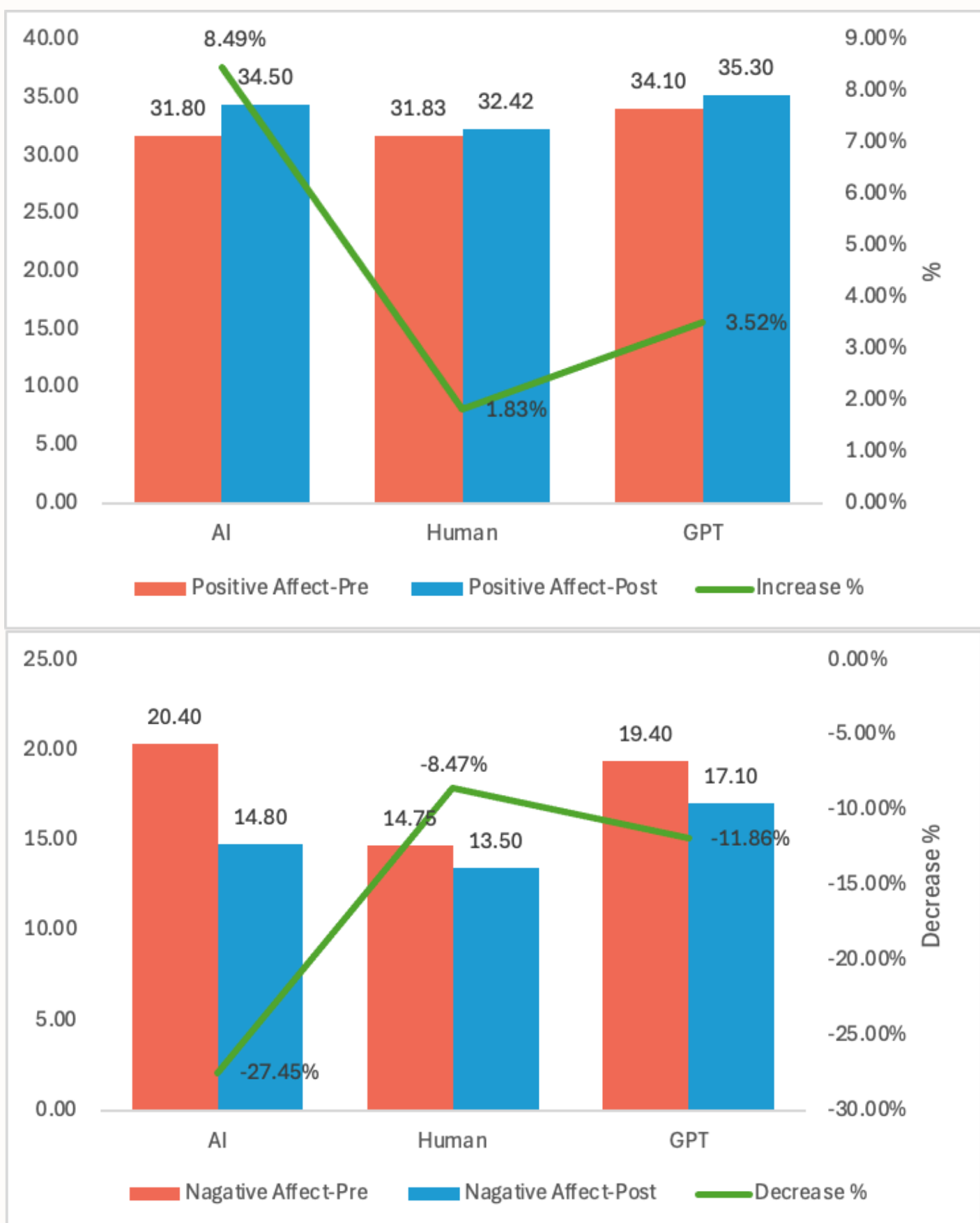
- Emotional Well-being: Positive and Negative Affect Schedule (PANAS) (Watson et al., 1988).
- Self-report Survey: 10 evaluation criteria

Qualitative Measures: 3 open-ended questions

The Comparative Study



Comparative Emotional Wellbeing Outcomes



AI coaching leads to the highest increase in positive emotions and the largest reduction in negative emotions

AI Coaching outperforms ChatGPT-4 across all dimensions
AI Coaching performs comparably to human coaching across all dimensions

Comparative Coaching Effectiveness Across 10 Dimensions

